

Mathematics — Units 1, 2, 4, 5 and 7

Revision Test 2026 | SSC-I (9th Grade) | Version 1 | 43-mark no-choice paper (24 MCQs)

Candidate: Seemab | Date: 19 August 2026 | Total Marks: 43 | Time allowed: 2 hours

41 / 43TOTAL SCORE
best paper in months**95.3%**

PERCENTAGE

A+

GRADE

23 / 24MCQ
95.8%

Syllabus audit: this paper was clean

Every one of the 24 MCQs was written against the Student Learning Outcomes printed in Seemab's own textbook, and the SLO citation for each appears in the answer key and in the subject's coverage tracker. **Each of the questions she got wrong was re-checked after marking and all sit squarely on a stated SLO**, so every deduction here is genuine. Nothing on this paper came from another subject's book or from outside her studied chapters. This is the first paper built under the rule adopted after Physics exam-13.

Section A — Multiple Choice Questions (24 × 1 = 24 Marks) — 23/24

#	Question (abbreviated)	Hers	Correct	Syllabus reference	Result
1	$\sqrt{2}$ is	B	B (irrational)	U1: real numbers as union of rationals and irrationals	CORRECT
2	The 8 in $\sqrt[3]{8}$ is called	B	B (radicand)	U1: radicals and radicands	CORRECT
3	Exponential form of $\sqrt[5]{x^3}$	A	A ($x^{3/5}$)	U1: transform radical to exponential form	CORRECT
4	0.333... is a	B	B (non-terminating recurring)	U1: terminating and non-terminating recurring decimals	CORRECT
5	$a^5 \times a^{-2}$	B	B (a^3)	U1: laws of exponents	CORRECT
6	45000 in standard form	B	B (4.5×10^4)	U2: standard form of scientific notation	CORRECT
7	$\log_2 8$	B	B (3)	U2: define logarithm to the base a	CORRECT
8	Integral part of a common logarithm	B	B (characteristic)	U2: characteristic and mantissa	CORRECT
9	Logarithm to base e	B	B (natural logarithm)	U2: common vs natural logarithm	CORRECT
10	$\log(mn)$ equals	B	B ($\log m + \log n$)	U2: four basic laws of logarithm	CORRECT
11	Factors of $a^2 - b^2$	B	B $((a+b)(a-b))$	U4: recall factorization of $a^2 - b^2$	CORRECT
12	$x^2 + 5x + 6$ factorizes to	A	A $((x+2)(x+3))$	U4: factorize $x^2 + px + q$	CORRECT
13	$a^3 + b^3$ factorizes to	C	A $((a+b)(a^2 - ab + b^2))$	U4: factorize $a^3 + b^3$	WRONG
14	HCF × LCM equals	B	B ($p \times q$)	U4: relationship between HCF and LCM	CORRECT
15	Factorizing $ka + kb + kc$	A	A $(k(a+b+c))$	U4: recall factorization of $ka + kb + kc$	CORRECT
16	$a^2 + 2ab + b^2$ is the expansion of	B	B $((a+b)^2)$	U4: recall factorization of $a^2 + 2ab + b^2$	CORRECT
17	Value of $ -7 $	B	B (7)	U5: define absolute value	CORRECT
18	Solution of $4x - 12 = 0$	A	A ($x = 3$)	U5: solve linear equation with rational coefficients	CORRECT
19	Exactly one of $a < b$, $a = b$, $a > b$ holds	C	C (trichotomy)	U5: properties of inequalities	CORRECT
20	Dividing $-2x < 6$ by -2	B	B ($x > -3$)	U5: multiplicative property; solve linear inequalities	CORRECT
21	Distance on a coordinate line	B	B ($ b - a $)	U7: distance between points on a coordinate line	CORRECT
22	Distance between (2,1) and (6,4)	A	A (5 units)	U7: find distance between two points	CORRECT
23	Mid-point of $(-3,5)$ and $(7,1)$	A	A $((2,3))$	U7: derive and use the mid-point formula	CORRECT
24	Points on the same straight line	B	B (collinear)	U7: collinear and non-collinear points	CORRECT

The single miss

Q13:

she chose

$$(a - b)(a^2 + ab + b^2)$$

, which is the factorization of

$$a^3 - b^3$$

, not $a^3 + b^3$. The two forms differ only in signs, and the rule is worth memorising as a pair:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

and

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

. The sign in the bracket is always *opposite* to the sign between the cubes. Notably she then applied the

correct

form in Section C, question 3(i)(b), so this was recall under pressure rather than a real gap.

Section B — Short Questions (3 × 3 = 9 Marks) — 8/9

Q	Topic	Awarded	Max	Examiner's note
2(i)	Radical to exponential form; laws of exponents	3	3	${}^4\sqrt{(x^5)} = x^{5/4}$ and ${}^3\sqrt{(a^2)} = a^{2/3}$, both shown through the intermediate bracket form. Then $x^8 \div x^4 = x^{8-4} = x^4$. Every step correct.
2(ii)	Laws of logarithms; $\log_5 125$	3	3	$\log m + \log n - \log p$ written correctly as log(mn/p) . Then $5^x = 125 = 5^3$, so x = 3 . Full marks.
2(iii)	Solving a linear inequality	2	3	The algebra is perfect: $7x \leq 28$, $x \leq 4$, and the solution set written properly as { x x ∈ ℝ ∧ x ≤ 4 } . The mark lost is for the second half of the question, which asked her to state which property of inequalities she used at each step . She showed the steps but never named the additive and multiplicative properties.

Section C — Long Questions (2 × 5 = 10 Marks) — 10/10

Q	Topic	Awarded	Max	Examiner's note
3(i)	Factorization; HCF and LCM	5	5	(a) $x^2 + 9x + 20$ split correctly by the middle term into $x(x+5) + 4(x+5) = (x+4)(x+5)$. (b) Recognised $8a^3 + 27b^3$ as $(2a)^3 + (3b)^3$, quoted the identity, and reached $(2a+3b)(4a^2 - 6ab + 9b^2)$. (c) HCF = 4ab , LCM = $24a^2b^3$, and the verification carried out both ways to $96a^3b^4$. Completely clean.
3(ii)	Distance and mid-point	5	5	Flawless. $AB = \sqrt{36 + 64} = \sqrt{100} = 10$. $M = (2, 6)$. Then $AM = \sqrt{9+16} = 5$ and $BM = \sqrt{9+16} = 5$, so M is equidistant from both. Formula, substitution and units all correct throughout.

What went well

Seemab's best paper in months and her highest MCQ percentage on any 24-question set.

Section C was perfect

, including a three-part factorization question and a full distance, mid-point and verification chain. She used the correct sum-of-cubes identity in Section C even though she had picked the wrong option for it in the MCQs, which shows the working knowledge is there.

Revision priorities from this paper

• The two cube identities as a matched pair.

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2); a^3 - b^3 = (a - b)(a^2 + ab + b^2). \text{ The middle sign flips.}$$

• Read the whole question.

The only other mark lost was for not naming the properties of inequalities, which the question asked for explicitly. When a question has two instructions, answer both.